CHAPTER 6

From 500 BCE to 500 CE

1. INTRODUCTION TO THE PERIOD (500 BCE – 500 CE)

A Turning Point in Indian Intellectual History

This millennium marks a major transformation in Indian mathematics.Shift from **ritual mathematics & calendrical calculations** to **systematic mathematical astronomy**.

The development of Indian mathematics between **500 BCE and 500 CE**, including the **geometry of the Sulba Sutras, influence of cultural exchanges, growth of astronomy, and major contributions of Aryabhata**, which laid the foundation for later mathematical progress in India.

Landmark Achievement

Culminates in **Aryabhatiya (499 CE)**

Written by **Aryabhata**

Introduced:

- Sine tables
- Planetary models
- Advanced astronomical calculations

CHARACTERISTICS OF THE PERIOD

- Continuous foreign invasions
- Cultural exchanges (Greek, Persian, Central Asian influences)
- Political unification and fragmentation
- Growth of universities like **Nalanda University**
- Expansion of trade routes and scholarly centres

ARYABHATA AND THE MATHEMATICAL REVOLUTION (499 CE)

Importance of Aryabhatiya

Aryabhatiya (499 CE) marks a **decisive turning point** in Indian mathematics.
Comparable in intellectual impact to **Ashtadhyayi** of **Panini** in linguistics.
Introduced:

- Mathematical rigor
- Systematic computation
- Astronomical precision

Key Mathematical Contributions

Plane Trigonometry – systematic sine (jya) tables.
Approximate solutions of **second-order difference equations** (for sine and cosine)
Use of innovative computational algorithms.
Called his results **"mathematical gems"** (Aryabhata's own expression)

Astronomy as a Driving Force

Before Aryabhata:

Mathematics mainly used for ritual and calendrical calculations.

After Aryabhata:

Study of planetary motion

Latitude and seasonal variation

Eclipses and conjunctions

Development of celestial coordinate systems

👉 Astronomy became **"geometry in motion"**

HELLENISTIC INFLUENCE AND CULTURAL EXCHANGE

Evidence suggests **Greek (Hellenistic) astronomy influenced Indian models**

Contact intensified after **Alexander's invasion (327 BCE)**

Exchange occurred through:

- Trade routes
- Diplomatic missions
- Scholars and philosophers

Northwest India (Gandhara) became a **gateway of knowledge transmission**

EARLY INVASIONS AND THEIR IMPACT

Persian Invasion – Cyrus the Great (6th Century BCE)

Occupied the **Upper Indus Basin**
Introduced **Aramaic writing system**
Led to development of **Kharosthi script**
Revival of written administrative traditions

Gandhara and Taxila

Taxila became a **cosmopolitan university town**
Hosted diverse intellectual traditions
Scholars like **Panini** likely worked here
Nursery of **Mahayana Buddhism** and scientific thought

Alexander's Campaign and Indo-Greek Relations
Alexander the Great (327 BCE)

Reached the Indus River
Began Indo-Greek scholarly interaction
Indo-Greek kingdoms flourished after his death

Cultural Outcomes

Gandharan (Indo-Greek) sculpture
Exchange in astronomy and philosophy
Long-term intellectual influence

Rise of Magadha – Intellectual Ferment

Religious & Philosophical Movements

Birthplace of:

- **Buddhism**
- **Jainism**

Emphasis on:

- Logic and debate
- Rational inquiry
- Ethical conduct

Encouraged **argumentation & logical reasoning** foundations for mathematics.

MAURYAN EMPIRE and ADMINISTRATIVE MATHEMATICS

Chandragupta Maurya
Founded first large Indian empire
Strengthened infrastructure and communication

Ashoka
Promoted Prakrit and Brahmi script
Encouraged knowledge exchange

Arthashastra by Kautilya
Advanced arithmetic knowledge
Loans and debts → early idea of **negative numbers (rṇa)**
Standardized units and measures
Binary & decimal usage in length/weight
Sexagesimal time likely adopted later via Greek contact

Shaka (Scythian) and Central Asian Influence

Entered India (1st century CE)
Important in evolution of numerals
Brahmi numerals developed through:
- Coinage
- Inscriptions

Gradual cultural assimilation

Gupta Era – The Golden Age

Political Stability

Gupta dynasty (4th–6th century CE) created strong governance

Capitals: **Pataliputra** and **Ujjain**

Ujjain as Scientific Hub

Became a **major astronomical center**

Zero meridian fixed through Ujjain

Attracted astronomers like **Varahamihira**

Siddhanta Texts

First formal **astronomical treatises**

Provided foundation for Aryabhata's work

Combined **Greek techniques with Indian innovations**

Huna Invasions and Intellectual Resilience

Mid-5th century invasions from Central Asia
Destruction especially in **Gandhara**
Targeted **Buddhist institutions**
Despite devastation, intellectual traditions **survived and adapted**

👉 Aryabhata's era coincided with this turbulence, showing **knowledge growth amid instability**

Overall Significance of the Millennium

Mathematical Developments

Ritual arithmetic → Abstract mathematics
Rise of trigonometry & algebraic thinking
Standardization of numerals

Cultural and Political Factors

Invasions brought scripts, arts, sciences
Empires enabled scholar mobility
Taxila and Ujjain became knowledge hubs

Legacy

Prepared ground for classical Indian mathematics
Established India as astronomical center
Cross-cultural interaction → innovation

Concluding Thought

"Mathematics does not grow in isolation."

Political change, trade, invasions, and cultural exchanges together led to the rise of **mathematical astronomy**, culminating in **Aryabhata** and his revolutionary work.

Chapter 6.2 – The Siddhantas and the Influence of Greek Astronomy

INTRODUCTION

In this lecture we will discuss an important phase in the history of Indian mathematics and astronomy known as the **Siddhanta period**.

This period marks a transition from traditional Vedic astronomical practices to a more systematic and mathematically structured astronomy.

The Siddhantas represent a blend of **indigenous Indian knowledge** and **Greek (Hellenistic) astronomical influence**, resulting in the emergence of mathematical astronomy in India.

Two major sources help us understand this period: the **Surya Siddhanta** and the **Pañcasiddhāntikā** of **Varahamihira**. Among these, the Pañcasiddhāntikā is considered more historically reliable for studying early developments.

Meaning of the Term "Siddhanta"

The word **Siddhanta** means *"established conclusion"* or *"settled doctrine."* In the context of astronomy, Siddhantas were not philosophical treatises but **practical manuals** that provided procedures for:

•Calculating planetary positions

•Predicting eclipses

•Determining conjunctions

•Timekeeping and calendar preparation

•These texts focused more on **methods and results** rather than detailed

theoretical explanations.

Major Sources of Siddhanta Knowledge

1. Surya Siddhanta

A well-known astronomical text.

However, it has undergone **many revisions over centuries**.

The version available today contains later additions.

Historically important because European scholars first encountered Indian trigonometry through this text in the 18th century.

2. Pañcasiddhāntikā – Varahamihira

Written around the **6th century CE**.

A **compilation of five earlier Siddhanta traditions**.

Serves as the primary source for understanding pre-Aryabhata astronomy.

Contains both **older material** and **Varahamihira's own contributions**.

The Five Siddhantas

Varahamihira's Pañcasiddhāntikā refers to five earlier astronomical systems:

Vasistha Siddhanta

Paitamaha Siddhanta

Romaka Siddhanta

Paulisa Siddhanta

Saura Siddhanta

Among these, **Romaka** and **Paulisa** show strong evidence of **Greek influence**, while the others are rooted more in **Indian traditions**.

Greek (Hellenistic) Influence on Indian Astronomy

•During the centuries following Alexander's invasion and Indo-Greek contact, Indian scholars came into interaction with **Greek astronomical ideas**, especially those developed in **Alexandria**. This influence led to several important transformations:

•Adoption of the **epicyclic model** of planetary motion

•Increased emphasis on **geometric modeling**

•Systematic astronomical calculations

•Greater precision in predicting celestial events

•Many historians believe that the work of **Claudius Ptolemy**, particularly his book *Almagest*, had an indirect impact on Indian astronomy.

Yavanapura and Ujjain

The Siddhanta texts mention **Yavanapura**, where “Yavana” refers to Greeks. This place is often associated with **Alexandria**. To compare astronomical observations made locally in India, scholars needed to understand the difference in longitude between Yavanapura and Indian cities.

Ujjain was chosen as the **prime meridian (zero longitude)** in Indian astronomy, highlighting India’s effort to adapt foreign knowledge into its own geographical framework.

Mathematical Techniques Introduced

Euclidean Algorithm
Used to simplify large fractions.
Based on repeated division and remainders.
Later evolved in India into Aryabhata's **Kuṭṭaka method** for solving equations.

Continued Fractions
Helped in approximating large ratios without loss of accuracy.
Particularly useful in astronomical calculations involving time cycles.

Sexagesimal System of Time
A notable foreign influence was the **sexagesimal (base-60) division of time**:
A day divided into 60 units, each further divided into 60.
Applied mainly to **time measurement** and **angles**.
All other arithmetic in India continued to use the **decimal system**.
This selective adoption shows a **blend rather than replacement** of indigenous methods

Indian Features Preserved in Siddhantas

Despite foreign influences, Siddhanta astronomy retained strong Indian characteristics:

Use of **decimal number system**

Sanskrit terminology for planets and constellations

References to Indian geography such as **Varanasi, Himalayas, Lanka**

Association with **Hindu rituals and calendars**

Religious timing of ceremonies based on astronomical calculations

Rise of Trigonometry

One of the most significant developments during this period was the emergence of **trigonometric concepts**:

Transition from **chord geometry** to **sine (Jyā / Ardhajyā)**.

Use of identities such as:

- $\sin^2\theta + \cos^2\theta = 1$
- Complementary angle relationships

Introduction of the **half-angle formula** for calculating sine values.

Sine Tables and Computational Advances

The circle was divided into **24 equal angular intervals**.

Step size often used was **$\pi/48$**.

Values were generated using:

- Half-angle formula
- Complementary angle rule

These tables laid the **foundation for systematic trigonometry** in India.

•**Aryabhata and Varahamihira – A Comparison**

•**Aryabhata** introduced a revolutionary approach using **sine differences** and mathematical equations.

•**Varahamihira** compiled earlier knowledge and contributed to interpretation and explanation.Aryabhata's work was more **innovative**, while Varahamihira's role was more **transitional and preservational**.

•**Astronomy and Astrology**

•During this era, **astronomy and astrology developed together**. While astronomy aimed at precise celestial prediction, astrology used these calculations for forecasting human events. Varahamihira himself was renowned both as an astronomer and astrologer.

Historical Significance

The Siddhanta period represents:
A **major transformation** in Indian scientific thought.
The birth of **mathematical modeling in astronomy**.
A successful **cross-cultural exchange of knowledge**.
The early development of **trigonometry and computational techniques** that influenced later scholars such as Brahmagupta and Bhaskara.

Conclusion

The Siddhantas served as a **bridge between ancient observational astronomy and classical mathematical astronomy** in India. Greek ideas influenced the framework, but Indian scholars expanded, refined, and integrated these concepts into their own intellectual tradition. This period ultimately prepared the ground for the groundbreaking contributions of **Aryabhata**, whose work established India as a major center of mathematical astronomy.

INTRODUCTION TO ARYABHATA

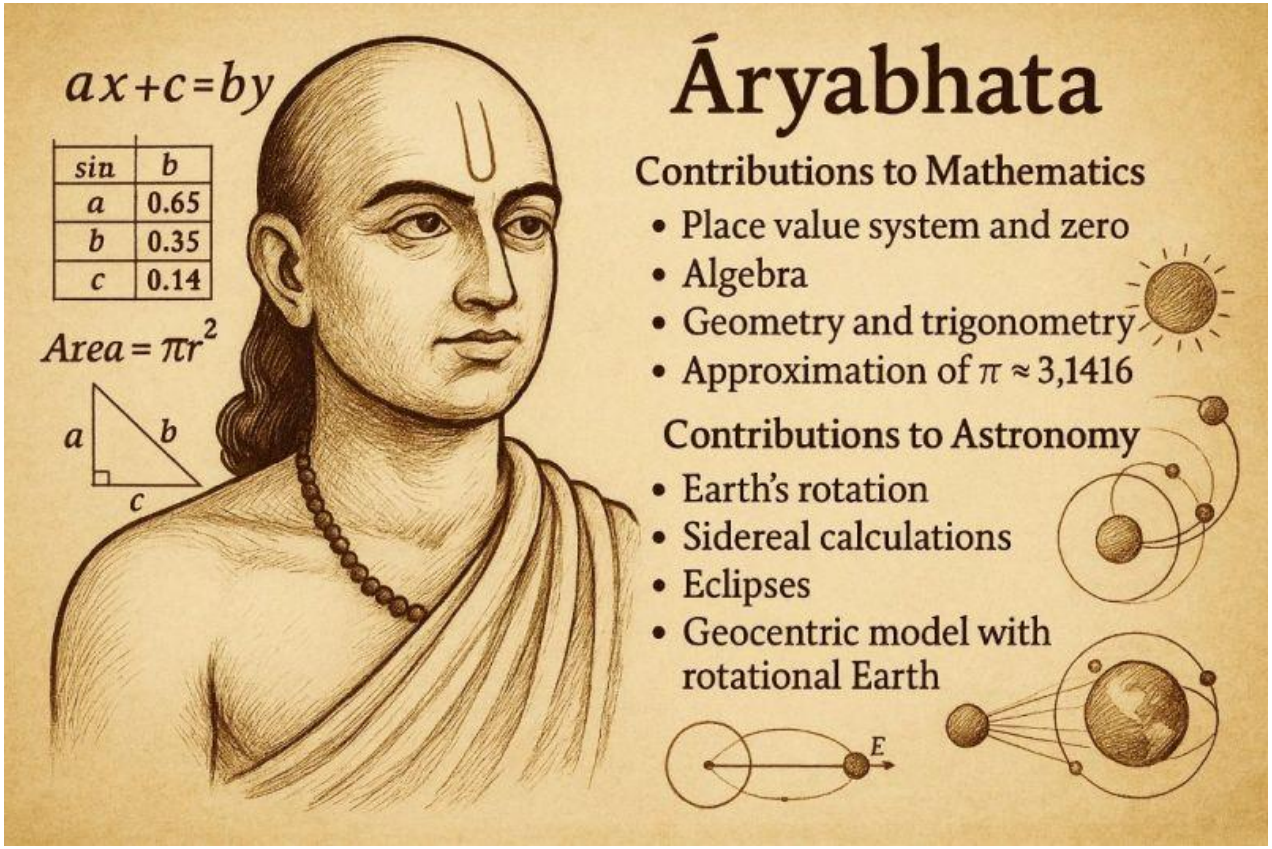
Aryabhata (born around **476 CE**) was one of the greatest **mathematicians and astronomers of ancient India**. His most famous work is the **Aryabhatiya**, written in **499 CE** when he was only **23 years old**.
This book marks a **major turning point in Indian mathematics and astronomy** because it combines several fields of knowledge, including:
Mathematics
Astronomy
Time calculation
Scientific reasoning

Aryabhata introduced many important scientific ideas, such as:
The **rotation of the Earth on its axis**, which explains day and night
Accurate calculations of planetary motion
Development of **trigonometry**
Use of **advanced algebraic methods**

Who Was Aryabhata?

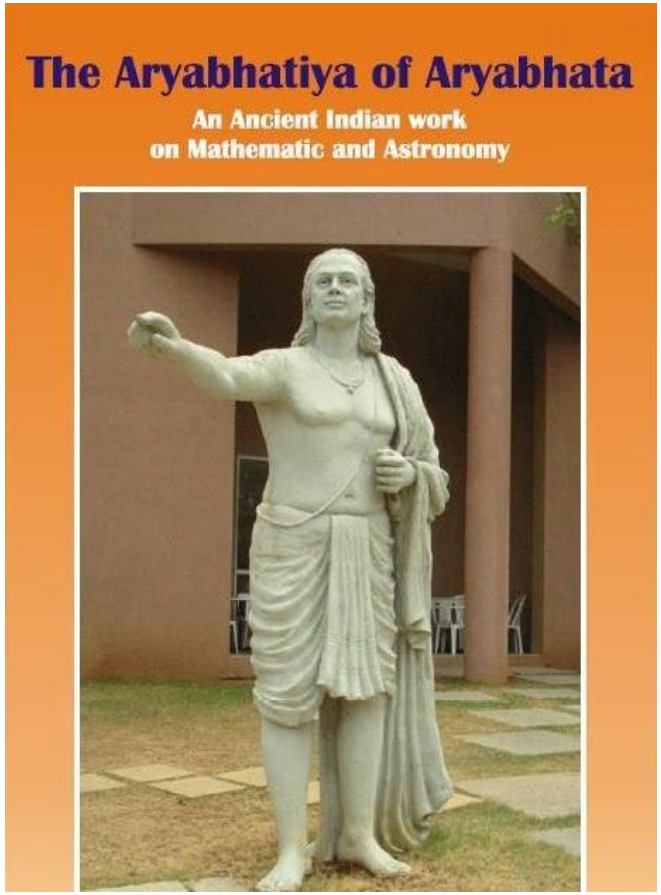
Aryabhata was a great Indian mathematician and astronomer who lived in **476 CE**. He lived in **Kusumapura** (modern Patna) and is believed to have been possibly born in the **Ashmaka** region.

He lived during the **Gupta Period**, a time known for great progress in science, culture, and education. This period also saw the rise of major learning centers such as **Nalanda University**.



•Aryabhata's religious identity is not clearly known. He was not specifically identified as **Hindu, Buddhist, or Jain**, but he invoked **"Brahma"** as a universal philosophical principle. His writings show **independent, rational, and scientific thinking** rather than strong religious influence.

•Aryabhata made important contributions to **mathematics and astronomy**, and his work played a major role in the development of scientific knowledge in ancient India.



Structure of the Aryabhatiya

The **Aryabhatiya** written by **Aryabhata** contains **121 Sanskrit verses** divided into **four chapters (Padas)**.

1. Gitikapada

Written in **Gitika meter** and contains **13 verses**.

Introduces important **astronomical constants and planetary revolutions**.

Defines the concept of **Yuga**, a large time cycle.

One Yuga = 4,320,000 years, after which planetary cycles repeat.

Serves as a **brief handbook for astronomers**.

A **Yuga = 4,320,000 years**

All planetary cycles repeat after one Yuga

Shows Aryabhata believed the **universe is periodic**

2. Ganitapada – Mathematics Section (33 Verses)

This is the **mathematical heart** of Aryabhatiya.

Deals with **mathematics**

Topics include:

Arithmetic operations

Square and cube roots

Geometry

Area and volume calculations

Progressions

Approximation of **π (pi)**

3. Kalakriyapada – Time Calculation (25 Verses)

Focuses on:

Measurement of time

Calendar calculations

Planetary periods

Philosophical idea: **Time has no beginning or end**

This was revolutionary because traditional belief linked time with mythology, but Aryabhata used **scientific reasoning**.

4. Golapada – Astronomy & Spherical Science (50 Verses)

Earth as a sphere
Planetary motions
Eclipses
Relative motion
Rotation of Earth

Famous Verse – Boat Analogy

Aryabhata explained **relative motion** using a boat example:
Just as objects on the shore seem to move backward when a boat moves forward, stars appear to move because Earth rotates.
This idea was **centuries ahead of Europe**.

Astronomical Ideas of Aryabhata

The astronomical ideas of **Aryabhata** were very advanced for his time and showed scientific thinking.

1. **Earth's Rotation-** Aryabhata explained that the **Earth rotates from west to east**, which causes **day and night**. This idea was very **controversial in ancient times**.

2. Relative Motion- He stated that **motion depends on the observer's point of view**, an early idea similar to the modern concept of **relative motion in physics**.

3. Size of the Earth- Aryabhata also **estimated the diameter of the Earth** and discussed the **height of the atmosphere**.

4. Explanation of Eclipses- He explained **solar and lunar eclipses scientifically using the shadow of the Earth and the Moon**, rejecting traditional **mythological explanations**.

In short: Aryabhata introduced **scientific and rational explanations for astronomical phenomena**, which was a major step in the development of astronomy in ancient India.

The mathematical contributions of Aryabhata:

- Formula for **area of a triangle**
- **Intersecting chords theorem** in geometry
- Description of the **decimal place-value system**
- **Kuttaka method** for solving indeterminate equations
- Solution of **linear equations**
- Accurate **approximation of π**

1. Area of a Triangle

Aryabhata gave the rule for the **area of a triangle**.
A = \frac{1}{2} \times \text{base} \times \text{height}
This formula means the area is **half the product of the base and height**.

2. Intersecting Chords Theorem

Another geometric result attributed to Aryabhata.
When a **chord of a circle and the diameter intersect at right angles**, the segments of the diameter satisfy:
AM×MB=MC2AM \times MB = MC^2AM×MB=MC2 This result comes from the **similarity of triangles formed inside the circle**.

3. Decimal Place-Value System

In Āryabhatīya, Aryabhata described a **positional number system**.
•Numbers were represented using **Sanskrit letters**.
•The word **"kha" (space)** represented an **empty place**.
•This system worked like the **modern decimal place-value system**, though zero was not yet treated as a number.
Example idea: 345
3 → hundreds
4 → tens
5 → units

4. Kuttaka Method (Pulverizer Method)

Aryabhata developed a method called **Kuttaka** to solve **indeterminate linear equations**.
Example equation mentioned : $ax+b=cy+d$

This can be rearranged as: $ax-cy=d-b$
The method uses repeated division similar to the **Euclidean algorithm** to find integer solutions.

5. Linear Equation Solution

Aryabhata solved equations of the form:

$ax+b=cx+d$

The solution given is

$x=(d-b)/(a-c)$

This is exactly the same as the modern algebraic solution.

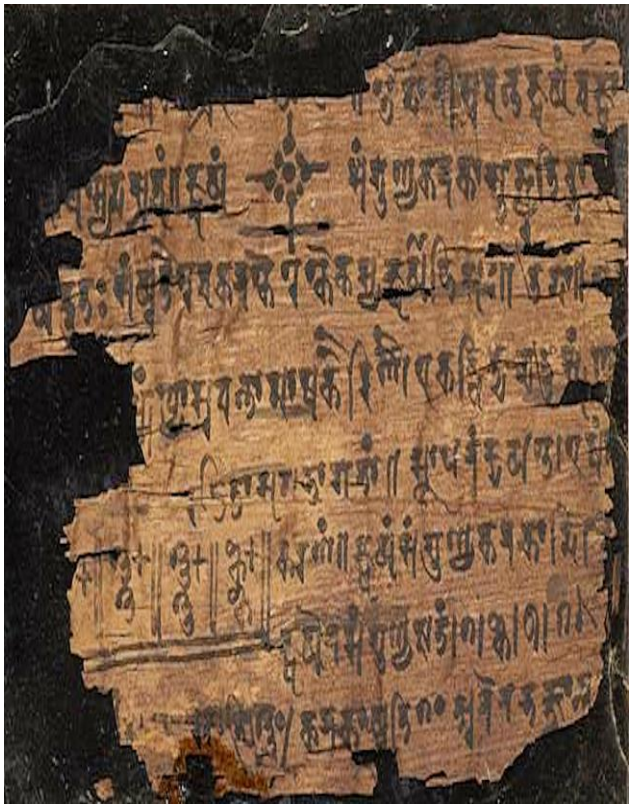
6. Approximation of π (Pi)

Aryabhata's famous approximation of π.

\pi \approx 3.1416

This value is **correct up to four decimal places**, which was extremely accurate for that time.

Bakhshali Manuscript



The Bakhshali Manuscript is an ancient Indian mathematical text written on birch bark and discovered in present-day Pakistan.

It is one of the earliest sources showing the development of **Indian mathematics and numerical methods**.

It contains arithmetic and algebraic rules, early use of the decimal system and zero, and methods for solving equations and approximating square roots.

Physical Features
Written on **birch bark** sheets.
Contains about **70 folios (pages)**.
The manuscript is currently preserved in the
University of Oxford.

Mathematical Importance

The manuscript is important because it shows early knowledge of:
Decimal number system
Symbol for zero represented by a **dot**
Algebraic symbols and arithmetic rules
Linear and quadratic equations
Series formulas
Square root approximation methods

Square Root Method- The manuscript gives an **iterative formula for approximating square roots**, which shows that ancient Indian mathematicians had **advanced algebraic thinking and numerical techniques**.

Date and Significance
The manuscript is generally estimated to belong to the period **300–500 CE**.
It is significant because it:
Shows **early symbolic mathematics**
Connects **Sulbasutra geometry** with later **algebraic developments**
Demonstrates **India's early use of the concept of zero**

Comparison: Aryabhatiya vs Bakhshali Manuscript

Feature	Aryabhatiya	Bakhshali Manuscript
Focus	Deals mainly with **astronomy and mathematics**	Focuses mainly on **arithmetic and algebra**
Style of Writing	Written in **Sanskrit verses (sutra style)**	Written as **practical mathematical problems with solutions**
Use of Zero	Explains the **concept of zero in a theoretical way**	Uses a **dot symbol to represent zero**
Algebra	Provides **general mathematical methods and rules**	Shows **applied algebra through solved problems**
Trigonometry	Contains **advanced trigonometric concepts and sine tables**	**Very little trigonometry** is present

Conclusion

The work of **Aryabhata** represents an important scientific development in ancient India. His famous book **Aryabhatiya** combined **mathematics and astronomy** and introduced many advanced ideas. He explained the **rotation of the Earth**, developed **trigonometry**, and gave a **scientific explanation of solar and lunar eclipses**. His work also contributed to **time measurement and rational scientific thinking**, which later influenced mathematicians like **Brahmagupta** and **Bhaskara II**.

The **Bakhshali Manuscript** from the same broad period shows the **practical side of Indian mathematics**. It contains arithmetic rules, algebraic methods, equations, and one of the earliest uses of the **symbol for zero** represented by a **dot**.
Together, these works prove that **ancient India had advanced scientific knowledge**, including:
A **developed decimal number system**
Algebraic methods and equations
Astronomical calculations and precision
Independent and rational scientific reasoning

“Aryabhata changed how India looked at the sky from mythology to mathematics.”

QUESTIONNAIRES

1. Explain the importance of Pañcasiddhāntikā as a historical source.(5)
2. What is meant by the term “Siddhanta”?
3. Why is the Surya Siddhanta not considered fully reliable for early astronomy?
4. Who wrote the Pañcasiddhāntikā?
5. Name the five Siddhantas mentioned by Varahamihira.
6. Discuss the Greek influence on Indian astronomy with examples.(5)
7. Which two Siddhantas show strong Greek (Hellenistic) influence?
8. Explain the use of the Euclidean algorithm in simplifying fractions.
9. Why was sexagesimal time division adopted only for time and angles?
10. What is the meaning of ardhajyā (jyā) in trigonometry?
11. Explain how astronomy became applied geometry during this period.
12. Who was Aryabhata? Briefly explain his contributions to mathematics.
13. Describe the structure of the Aryabhatiya and explain its four chapters.
14. What is the Kuttaka (Pulverizer) method developed by Aryabhata? Explain its purpose.
15. Explain Aryabhata’s contribution to the decimal place-value system.
16. Discuss Aryabhata’s contribution to trigonometry and sine tables.
17. Explain Aryabhata’s method for solving linear equations.
18. Explain Aryabhata’s ideas about Earth’s rotation and day-night cycle.
19. Discuss Aryabhata’s explanation of solar and lunar eclipses.
20. Discuss the relationship between astronomy and mathematics in Aryabhata’s work.